TIER 1 — edge controllers, one active and one standby per floor

Raspberry Pi 400 (ARM64) — actives

- edge-level0-active
- edge-level1-active
- edge-level2-active

amd64 worker — standbys (same image and spec)

- edge-level0-standby
- edge-level1-standby
- edge-level2-standby

PUT door state (own floor)

BuildSim
3D viewer · 124 fire doors
sprinklers · floor graph

MQTT broker (Mosquitto)

edge → server: alerts · decisions · heartbeat
server → edge: readings · commands

Every 5 s per floor: rule + isolation forest, own doors released in the same tick.
Heartbeat 10 s; standby promotes after 30 s.
Works without server, broker or LLM.

floor plans door graph

effects · cards · doors of a taken-over floor

TIER 2 — fds-server

physics engine + sensor gateway
956 spaces · spread along door graph
sensor faults (noise, drift, stuck)

agent
alert queue (3 parallel) → LLM verdict → coordinate floors → watch / stand-down → ALL CLEAR

pipeline
MQTT → TimescaleDB

control API
fire · false alarm
viz/clear · status

LLM inference
LiteLLM → SGLang, Qwen2.5-14B (GPU)
verdict p50 2.1 s · p95 3.2 s

TimescaleDB
sensor_readings · edge_decisions
heartbeats · llm_overrides

Grafana · Phoenix · ClearML
retrain → detector.onnx → deploy.sh edge